Sachin A

+91 63796 40407 asachin1966@gmail.com linkedin.com/in/sachin-a github.com/Sachin1966

Professional Summary

Aspiring AI Developer with hands-on experience in Machine Learning, Deep Learning, Generative AI, and Retrieval-Augmented Generation (RAG). Skilled in building intelligent systems using Python, PyTorch, TensorFlow, Scikit-learn, and LLM frameworks, with practical experience in NLP, time-series modeling, computer vision, and AI-powered applications through real-world projects.

Education

- **Sri Shakthi Institute of Engineering and Technology** — Coimbatore, India
  *B.Tech – Artificial Intelligence and Machine Learning — CGPA: 7.47** — *2023 - 2027*
- **Little Flower Matriculation Higher Secondary School** — Karur, India
  Higher Secondary Certificate (HSC) — Percentage: 82% — *2022 - 2023*

Skills

- **Programming:** Python, Object-oriented Programming
- **Backend Frameworks:** Flask, FastAPI, REST APIs
- **Databases:** MySQL, MongoDB, FAISS, Pinecone, ChromaDB
- **Tools:** Git, GitHub, Docker, Huggingface Spaces
- **AI / ML:** Machine Learning, Deep Learning, Generative AI, Agentic AI, LangChain, RAG, Prompt Engineering
- **ML & Data Analysis Libraries:** NumPy, Pandas, Matplotlib, Seaborn, Plotly, Scikit-learn, PyTorch, TensorFlow
- **Frontend:** HTML, CSS, JavaScript, React
- **Soft Skills:** Problem Solving, Communication, Team Collaboration, Continuous Learning

Projects

- **ContextEngine AI Document Intelligence System**
  - **Objective:** Develop an AI-powered document intelligence system for context-aware knowledge retrieval and question answering.
  - **Implementation:** Built a Retrieval-Augmented Generation (RAG) pipeline with semantic embeddings and vector search, indexing 100+ documents and retrieving source-grounded answers using LLMs.
  - **Impact:** Enhanced information discovery and reduced hallucinations by providing accurate, citation-backed responses from large document collections.
  - **Tech:** Python, LangChain, ChromaDB, RAG, NLP, Embeddings, Vector Search, LLMs
  - **Deployment:** *Deployed on Hugging Face Spaces*
  - **GitHub:** https://github.com/Sachin1966/Context-Engine-
- **Autonomous Telemetry Learning and Generation Agent**
  - **Objective:** Develop an AI system to learn complex telemetry patterns and generate realistic synthetic time-series data.
  - **Implementation:** Trained an LSTM-based Variational Autoencoder (VAE) in PyTorch to capture temporal dependencies and autonomously synthesize high-fidelity telemetry sequences.
  - **Impact:** Improved AI-driven simulation, anomaly analysis, and model validation by generating diverse synthetic datasets with minimal human intervention.
  - **Tech:** Python, PyTorch, LSTM, Variational Autoencoder (VAE), Deep Learning, Time-Series Modeling.
  - **GitHub:** https://github.com/Sachin1966/Telemetry-Agent
- **AI Creative Studio Platform**
  - **Objective:** To develop a premium AI-powered creative studio that enables users to generate, edit, analyze, and manage images, scripts, audio, and prompts through an integrated multi-modal web platform.
  - **Implementation:** Built 4 core modules — image generation, script creation, audio processing, and vision analysis — using FastAPI, React, WebAssembly, and browser-based processing.
  - **Impact:** Enhances creativity and productivity by providing fast, AI-powered content creation and editing in a single platform.
  - **Tech:** FastAPI, React, AI APIs, WebAssembly (WASM), and Web Audio API.
  - **GitHub:** https://github.com/Sachin1966/AI-Creative-Studio

Achievements

- **TATA National Quiz:** Participated in a national-level competitive quiz demonstrating analytical and problem-solving skills.
- **ISRO Hackathon:** Participated in a collaborative hackathon focused on real-world space and engineering challenges.

Technical Certifications

- **Prompt Engineering — AWS — 4-Week Course (March 2025)**
- **Fundamentals of Generative AI — Databricks Academy— 4-Week Course (June 2026)**
- **Building Single Agent Applicattion — Databricks Academy — 4-Week Course (June 2026)**